Electronics 1: Overview

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SAGI 2025

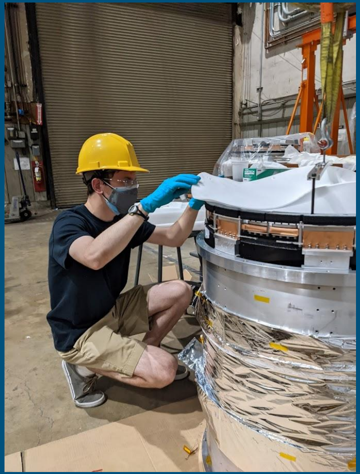
Goals of Electronics Lectures Series

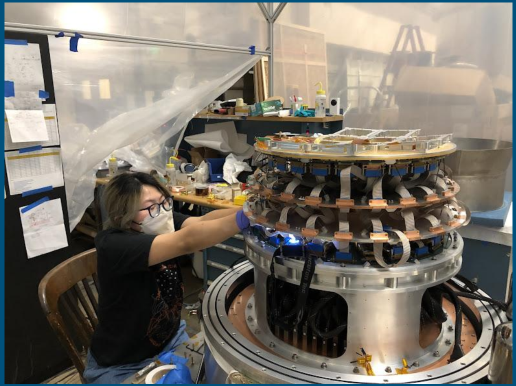
- To fully internalize topics covered usually takes a full semester course. We have 12 hours total!
- The goal is **not** to fully understand electronics concepts shown. It is to give you exposure and intuition for the most important considerations to <u>make a highly sensitive measurement.</u>
- I will be light on derivations, and will instead place an emphasis on real world examples and applications, and try to justify to you the importance of understanding the concepts and techniques I describe.

Electronics/Instrumentation in the Field

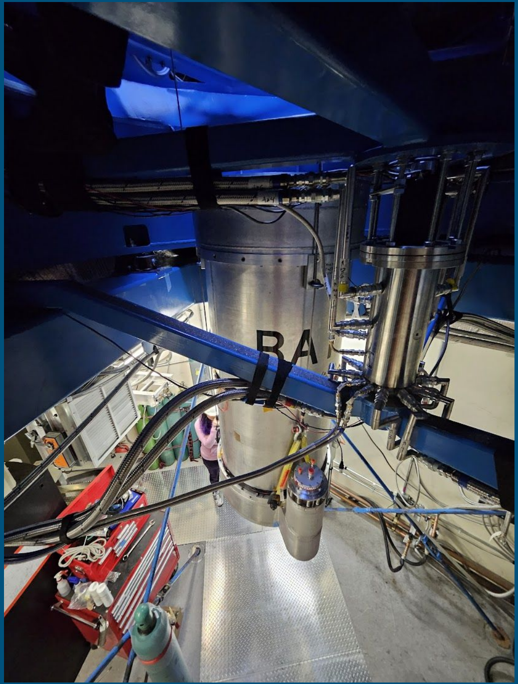
BA2 2020-2022 at High Bay

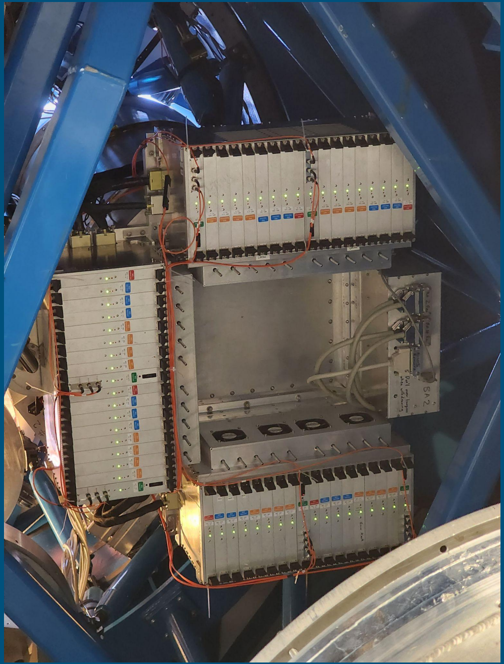
- 5 total cold runs w/ modules over 2 years
- Optics, thin window, curved focal plane, readout electronics, housekeeping electronics all gradually incorporated
- Numerous module and receiver calibrations undertaken in preparation for deployment
- Receiver packed and shipped to South Pole in Sept. 2022





BA2 2022 Deployment - Installation





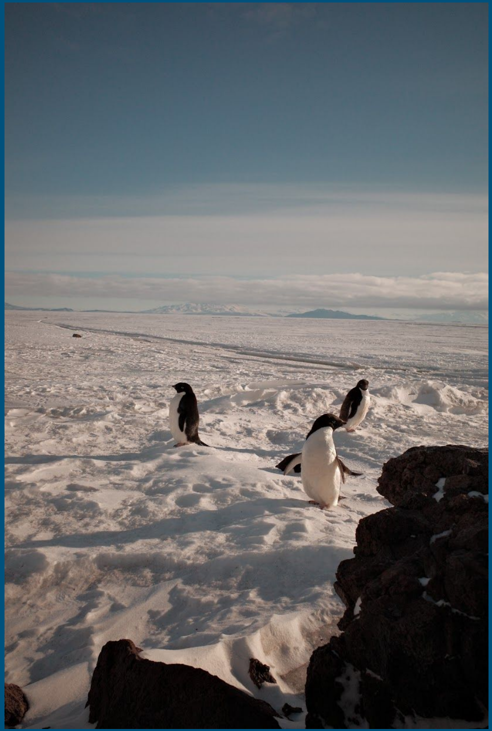
BICEP Array - 2023/2024 Deployment

- Extremely small team - 5 total for full season/3.5 months

- Summer season consists of maintenance, upgrades, calibrations, small dumpster fires etc.

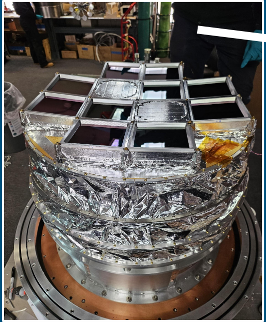
- Deployed **8 new modules** into the field between BA1 (30/40 GHz) and BA2 (150 GHz)

BICEP Array - 2023/2024 Deployment

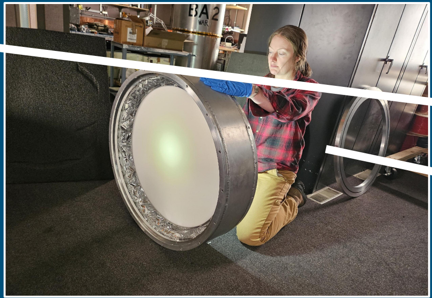


BICEP Array - 2023/2024 Deployment

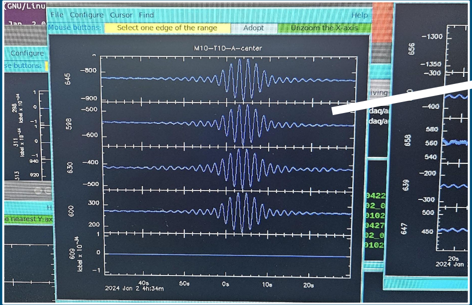
Upgrades



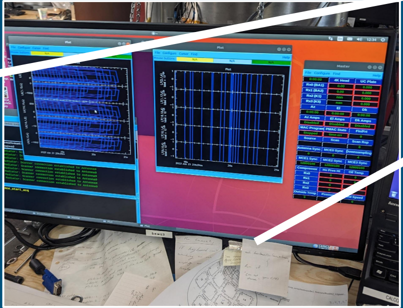
Maintenance

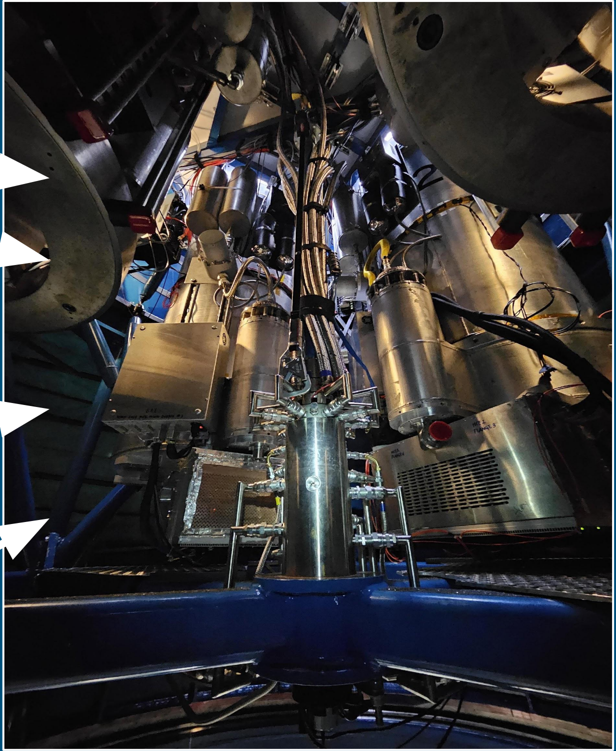


Calibration



Everything else...



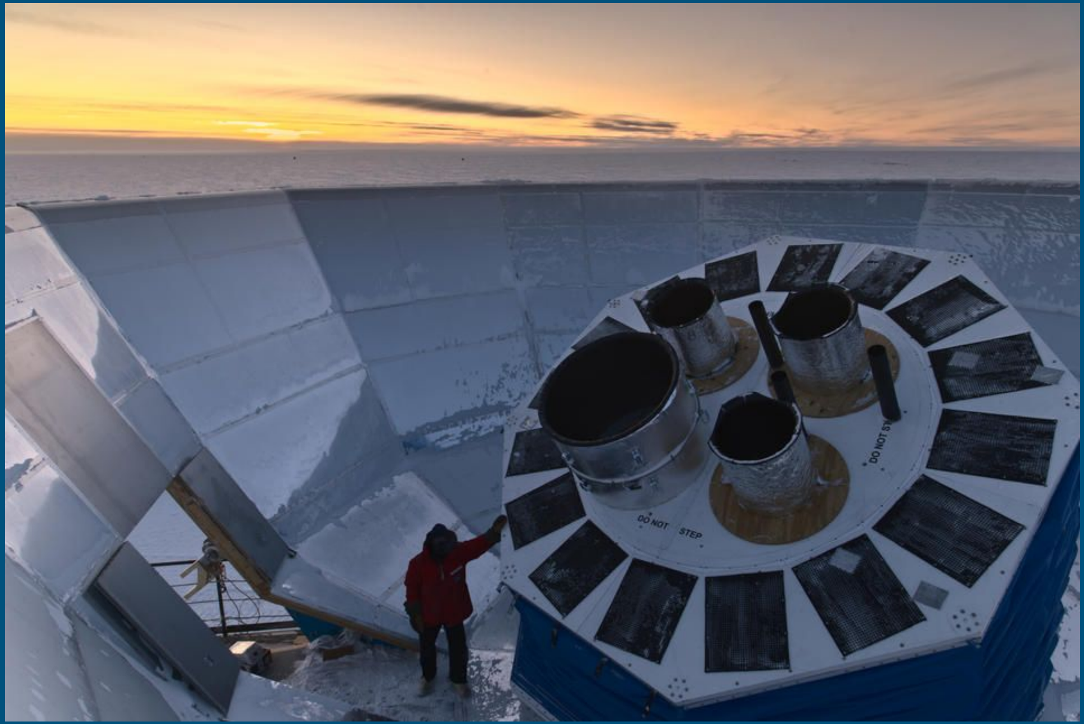


What aspects of ‘electronics’ aid us in making measurements?

1. Detectors - antennas, bolometers, CCDs, calorimeters, spectrometers…
2. Noise - photon noise, noise temperature, cross talk
3. Amplification - Op-amps, LNAs, SQUIDs
4. Digitization - ADCs, DACs, multiplexing
5. Signal processing - filtering, fitting, fourier transforms

BICEP Array - Precise Measurement in Practice

- CMB telescope placing strongest constraints on B-mode polarization of CMB by inflationary gravitational waves
- 4 telescopes measuring at multiple frequencies
- Located at South Pole
- Can observe same sky patch 24/7
- Small telescope -> cheaper, easier to constrain systematics



Credit: Nathan Precup

Detectors: Examples

What are some examples of detectors you can think of?

In this room?

In your day to day life?

In medicine?

In physics?

Detectors: Performance metrics

What are some key metrics that might tell you how well a detector might perform?

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What are some key metrics that might tell you how well a detector might perform?

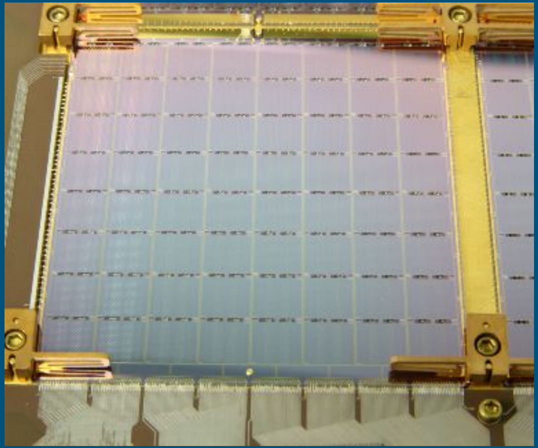
Responsivity: How efficiently an input is converted to an output

Noise: Detectors have intrinsic noise, background noise. What is the minimum detectable signal level?

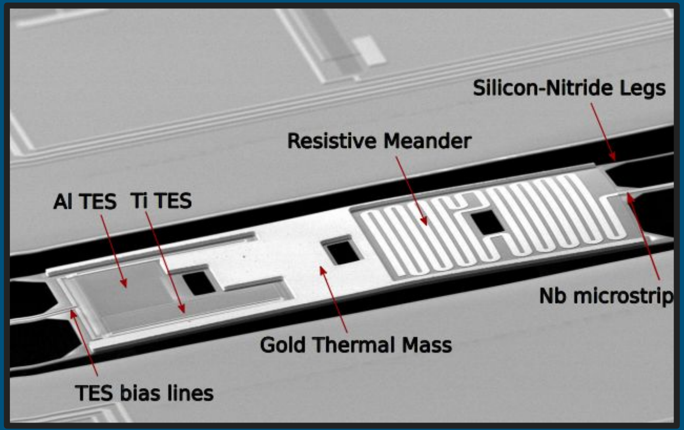
Coupling: How well can a signal be transferred between devices ie. amplification

Dynamic range: How small and how big a signal can be detected? Can this device be ‘maxed out’?

BICEP Array - Detectors: Transition Edge Sensors


Wafer Detector Tile


Slotted Antenna Array



Transition Edge Sensor

Noise: Examples

What are some types of noise you are aware of that would matter in electronics?

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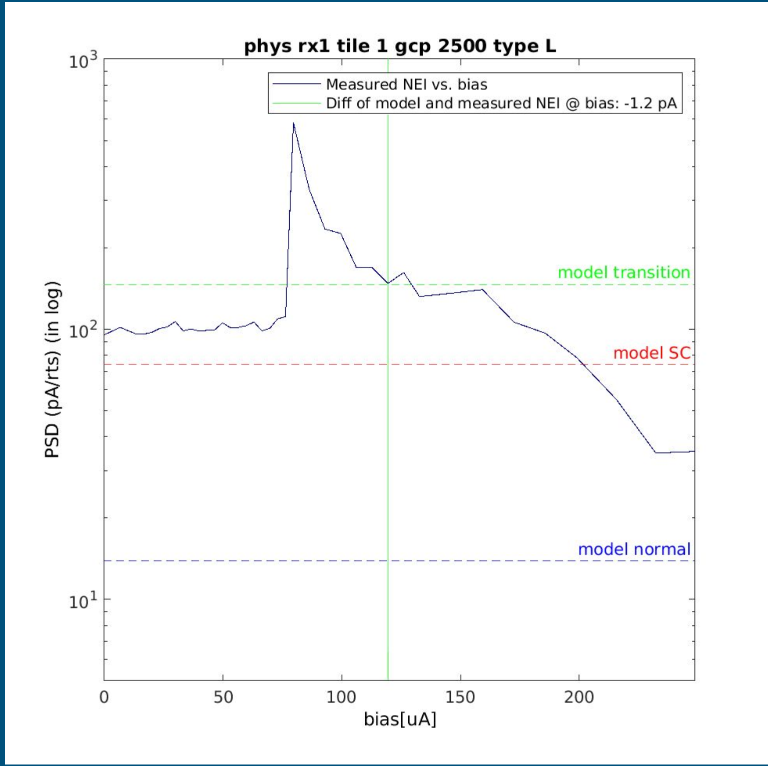
What are some types of noise you are aware of that would matter in electronics?

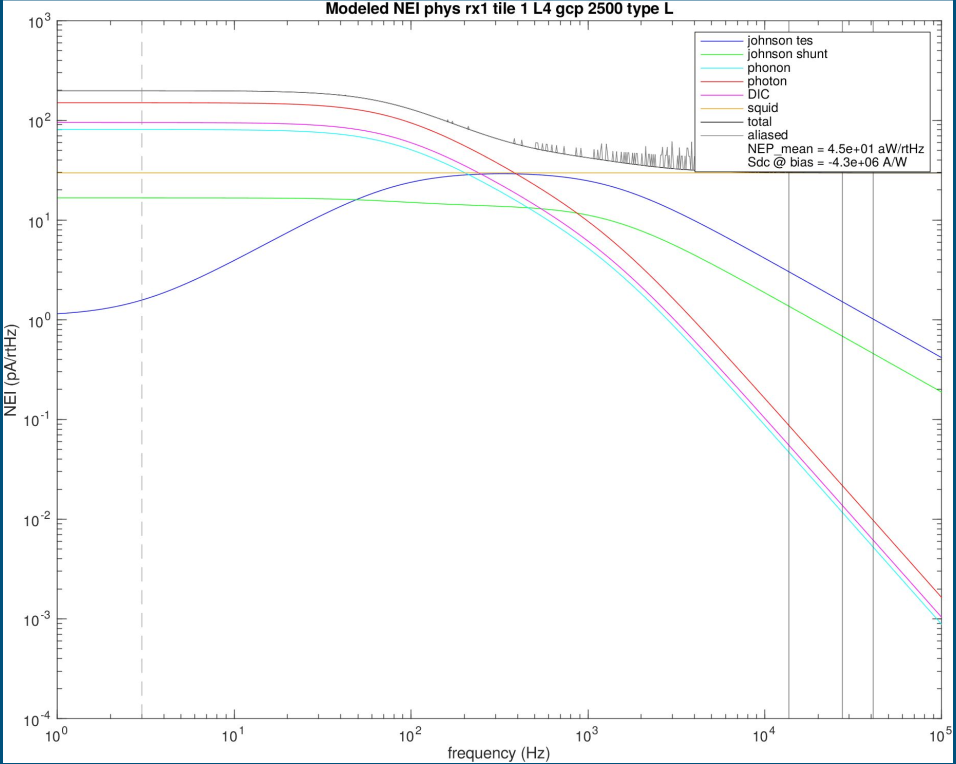
Johnson/thermal noise: Noise generated by brownian motion of particles. **How does one mitigate?**

Photon noise: Intrinsic noise associated with photon detection.

Flicker or 1/f noise: Present in semiconductor materials, usually due to contamination, material stressors

BICEP Array - Noise





Amplification

What would one consider when amplifying a signal?

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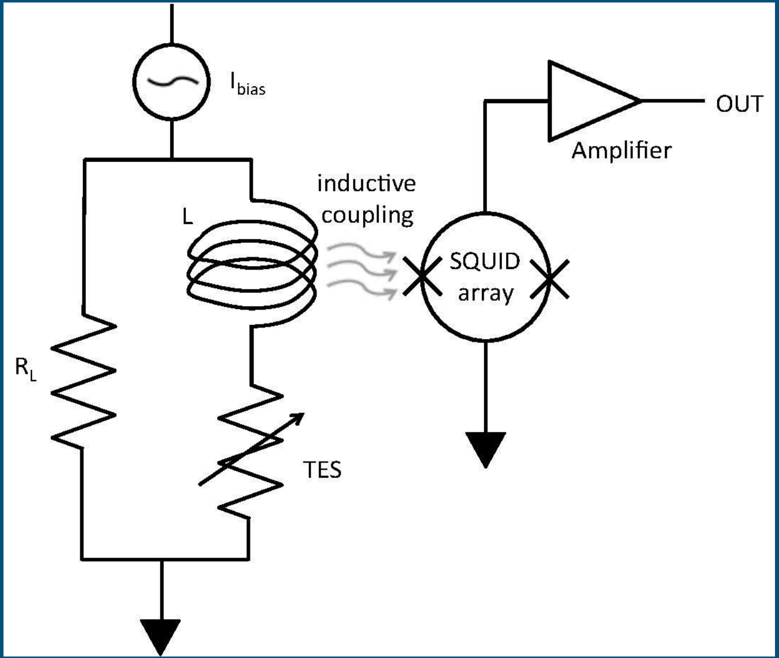
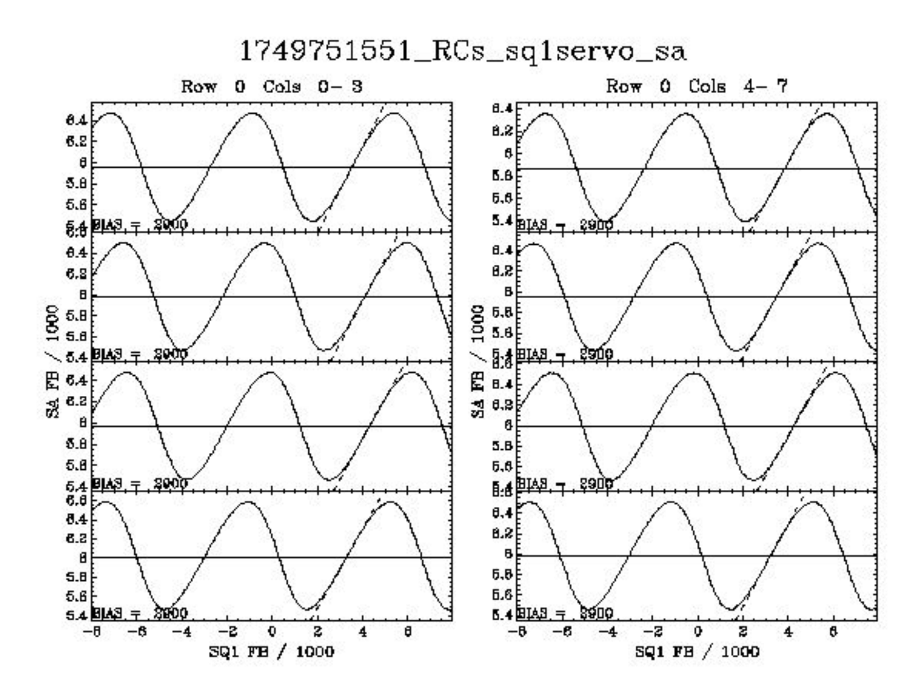
Frequency response: Are all frequencies of a signal amplified equally?

Noise: Does an amplifier add additional noise to the signal?

Stability: Will this amplifier work consistently in a variety of conditions/loads?

BICEP Array - Amplification: SQUIDs





Digitization

Digitization involves recording the recording of a signal. What would one consider?

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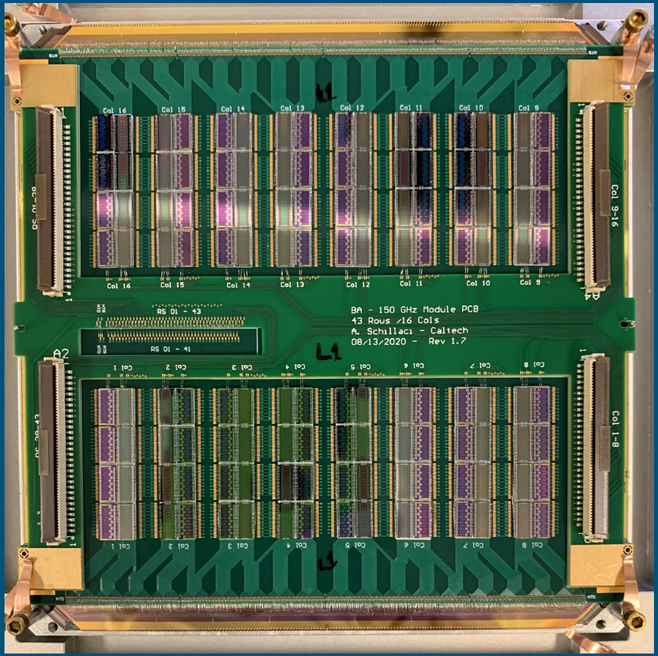
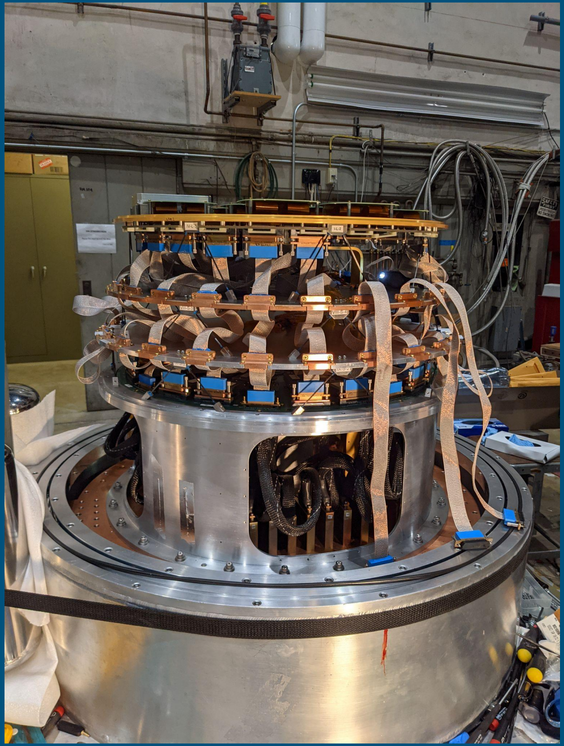
Sampling rate: How fast can you sample a signal per second? What resolution?

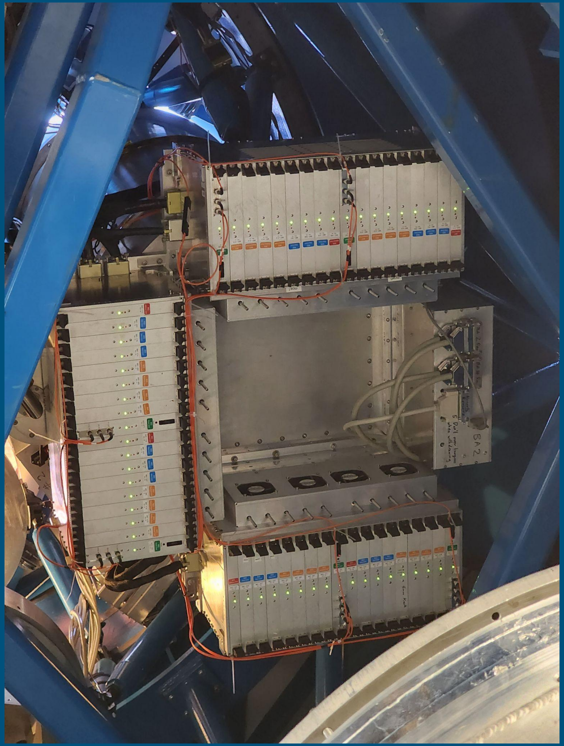
Data volume: Can I record the data fast enough to not lose information? How much storage do I need?

Multiplexing: What if I have 1000s of detectors to record? Can I record them all at the same time or do I have to use strategies to observe more detectors than I can record simultaneously?

BICEP Array - Digitization







Signal Processing

Again, what are some consideration when I need to to ‘process’ an electronic signal?

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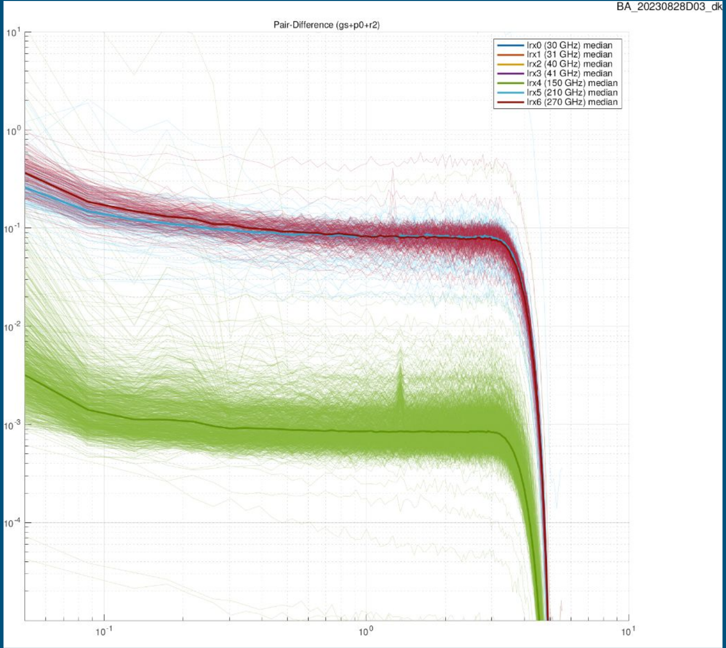
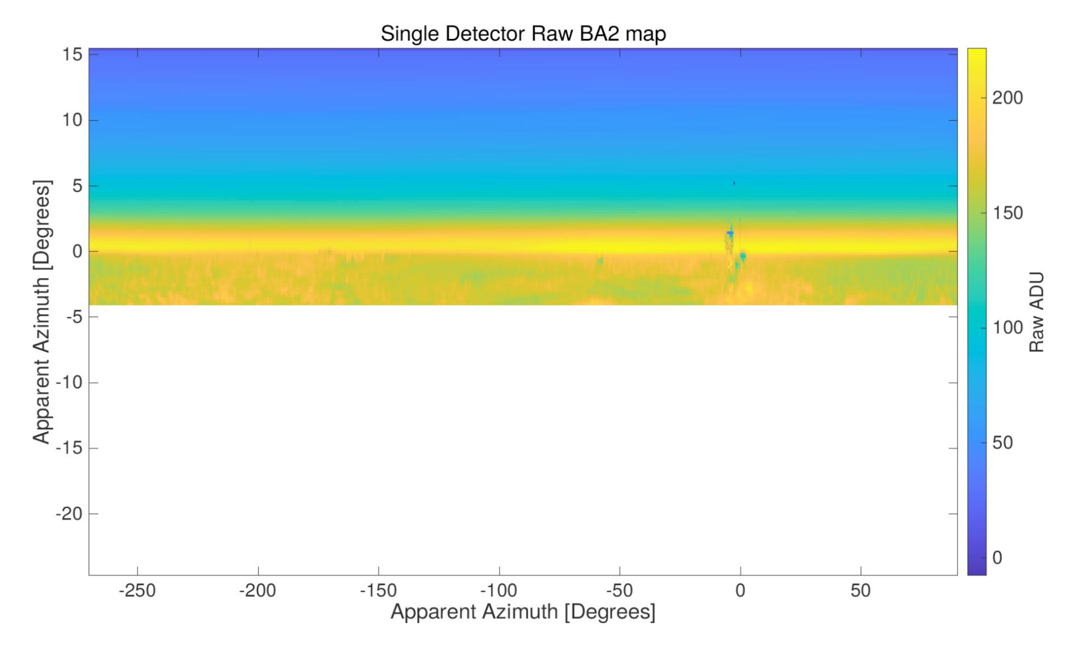
Filtering: Is some of the frequency content unwanted? Can I remove?

Fitting: Are there fluctuations in time that I can fit and subtract out?

BICEP Array - Signal Processing





Topics we will cover

Passive filters

Fourier transforms of analog signals

Nonlinear devices

Amplification and gain

Physics of Noise

Noise in Electronic Systems

Dynamic range of systems

Signal processing techniques

Frequency conversion systems

Multiplexing

Any other topics you want to hear about?

Any questions?